



IDENTIFYING CRITICAL IN-WATER HABITAT FOR LEATHERBACK SEA TURTLES (*DERMOCHELYS CORIACEA*) NESTING AT SANDY POINT NWR, ST. CROIX USVI

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INTRODUCTION

- Animals move in response to **environmental factors** (predators, prey, ocean circulation).
- Leatherback turtles (*Dermochelys coriacea*), are **highly migratory** and are listed as **Vulnerable** worldwide (Wallace et al. 2013).
- Northwest Atlantic leatherback nesting numbers have been **declining since 1990** (Northwest Atlantic Working Group 2018).

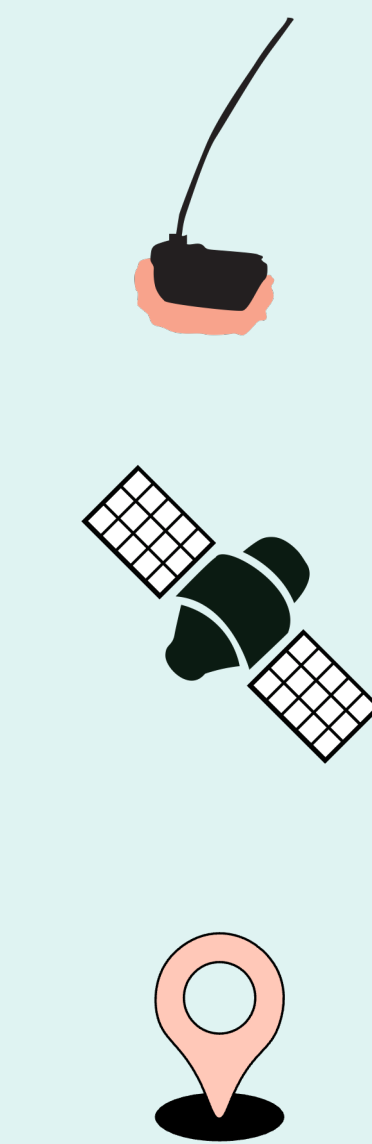


Figure 1: Nesting leatherback sea turtle at Sandy Point. Photo: Jeremy Smith. * Photograph taken during the Leatherback Project at SPNWR and used with permission - U.S. Fish and Wildlife Service.

STUDY OBJECTIVE:

The goal of this study was to **identify important in-water habitats** for leatherback sea turtles during the breeding season. Knowledge of critical habitats can inform the location of protected areas.

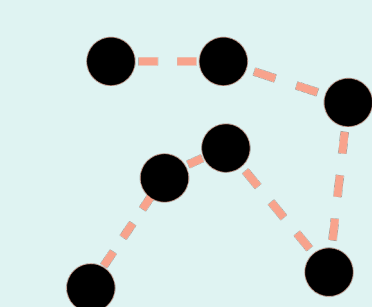
METHODS



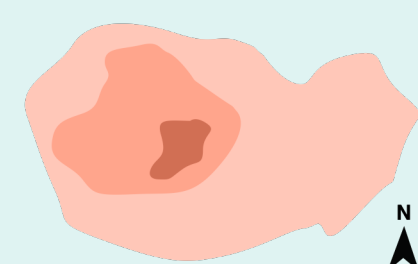
1) Outfitted 10 nesting female leatherbacks with **SPLASH10 satellite tags** in May 2020 and 2021 at Sandy Point NWR, St. Croix USVI using direct attachment methods (Dodge et al. 2014).

2) Tags communicated with Argos satellites to **transmit data**.

3) Downloaded satellite-transmitted **location data** from the tagged turtles via the Wildlife Computers data portal.



4) **Filtered and smoothed** the data to remove location errors and regularize the time interval between points (3 hours) using *foieGras* package in R.



5) Created a density map using **Kernel Density** approach in ArcGIS Pro.

Figure 2: Overview of study methods.

RESULTS

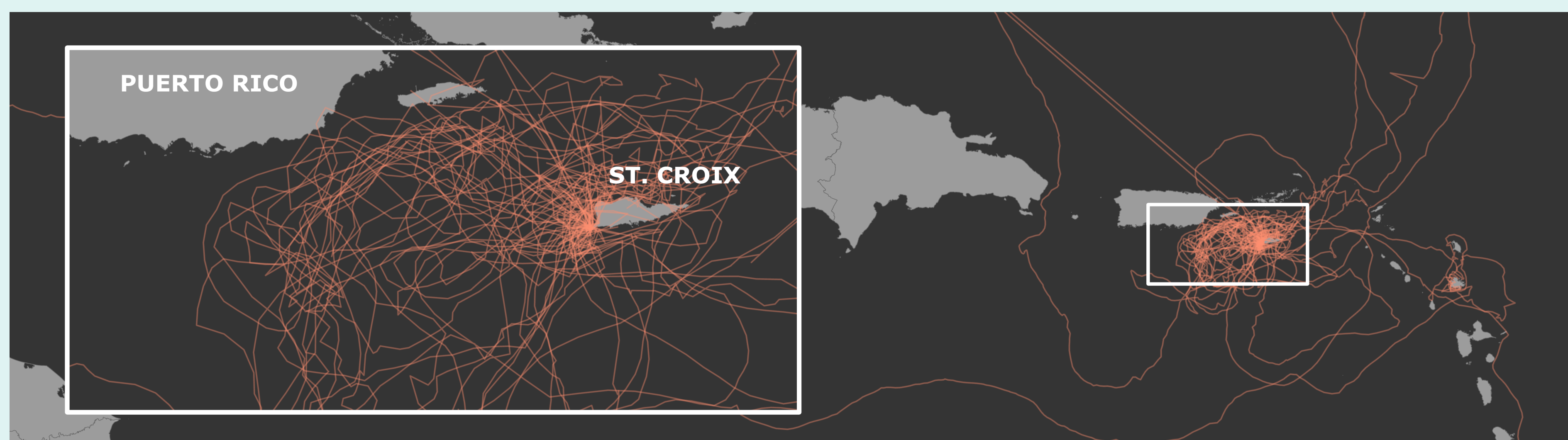


Figure 3: Map of leatherback tracks near St. Croix (inset map) and around the wider Caribbean region.

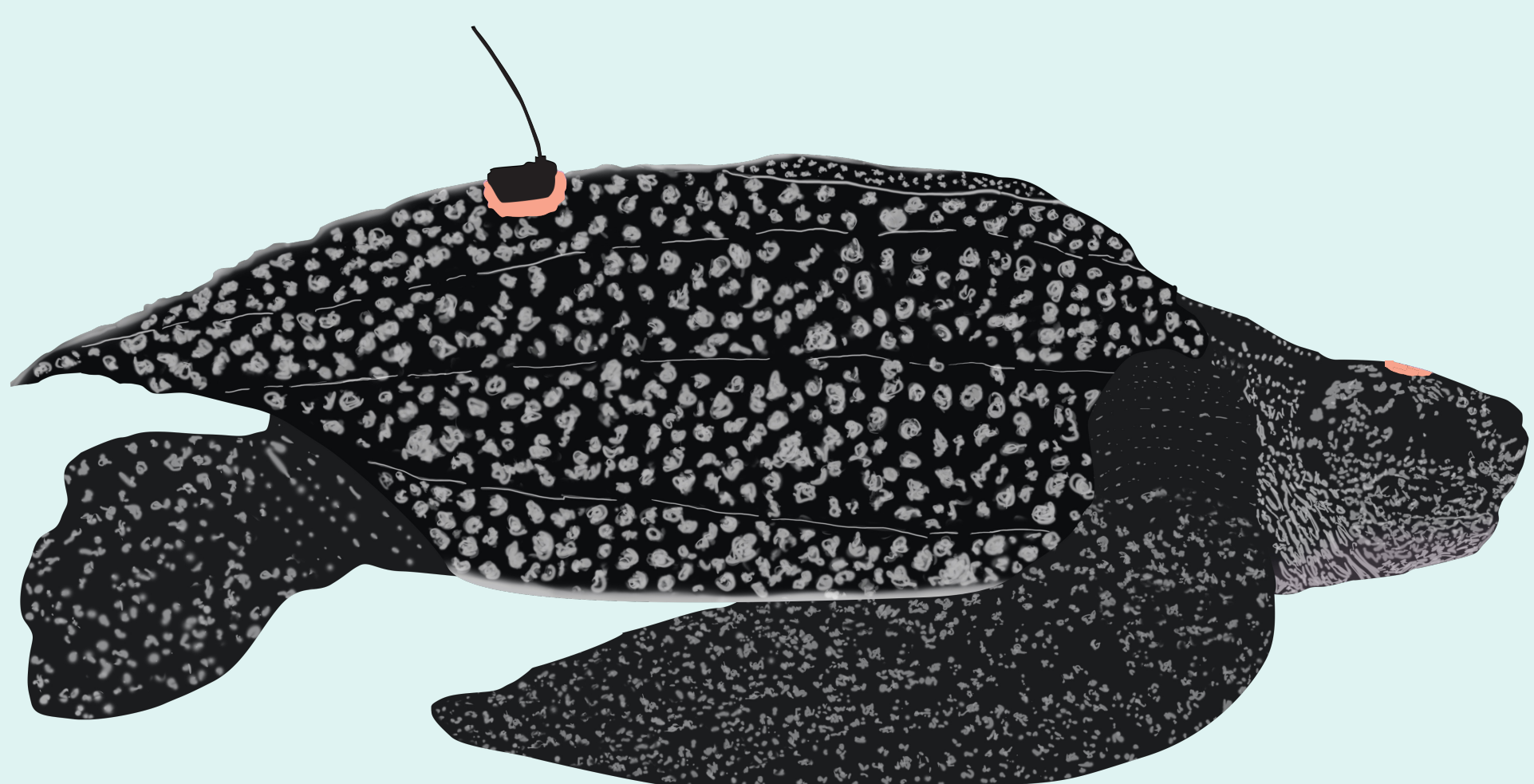


Figure 4: Leatherback turtle outfitted with SPLASH10 satellite tag.

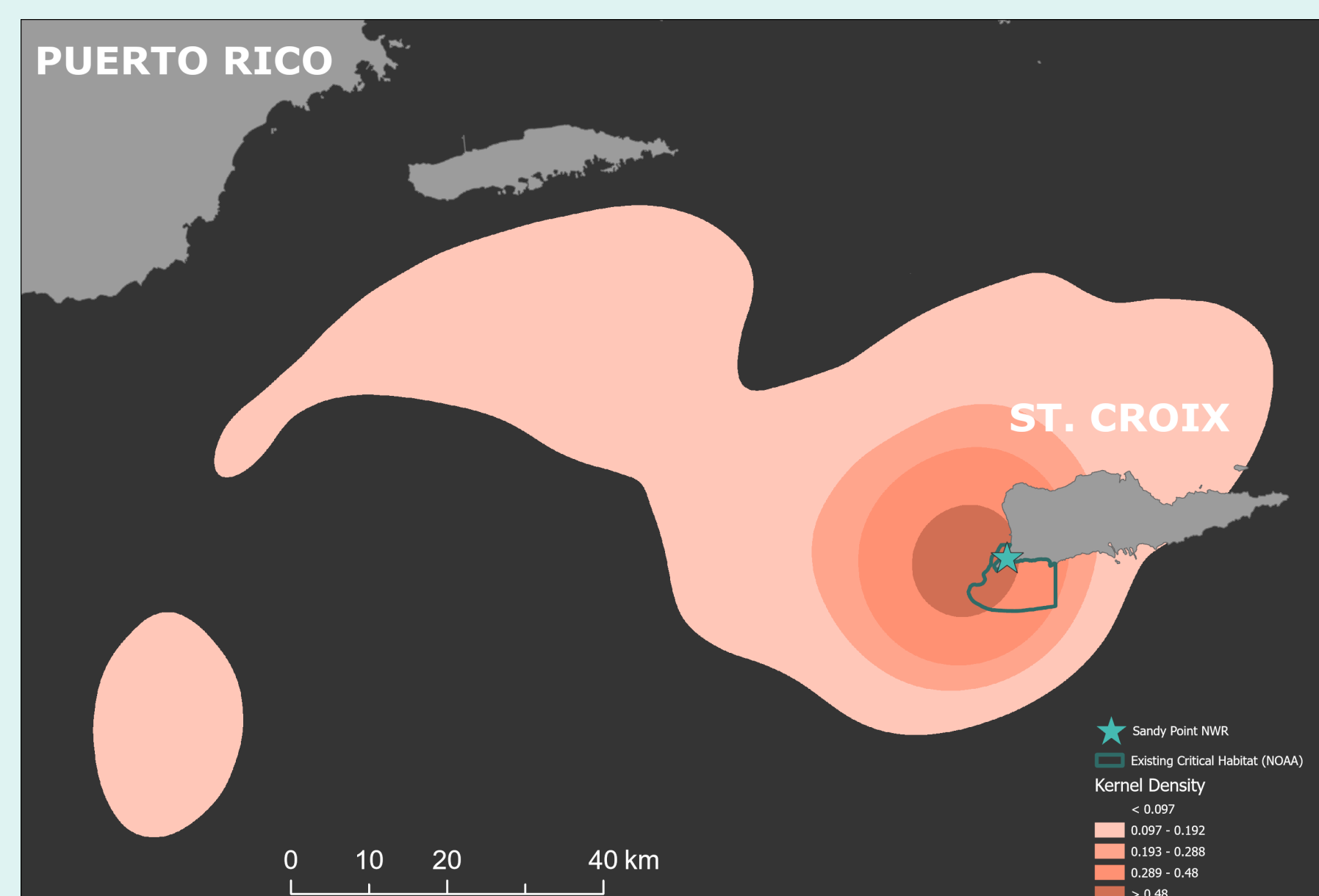


Figure 5: High-use areas of leatherbacks during interesting periods defined using Kernel Density analysis. Warmer colors represent greater estimates, indicating highest use.

DISCUSSION

Important in-water habitats (high-density areas) are located:

- in the offshore waters surrounding Sandy Point (within < 20 km) and St. Croix
 - Depths ranging from 0 – 4,000 m
- throughout the deep (depths > 4,500 m) trench between Puerto Rico and St. Croix

These density maps suggest that - to protect in-water areas that are important to vulnerable nesting leatherbacks - **existing Critical Habitat area should be expanded**, and an **additional protected area** should be designated in the Virgin Islands trench.

FUTURE WORK:

- Track more turtles!*
- Characterize behaviors*
- Identify environmental properties of important habitats*

REFERENCES

- Dodge KL, Galuardi B, Miller TJ, Lutcavage ME (2014) Leatherback turtle movements, dive behavior, and habitat characteristics in ecoregions of the Northwest Atlantic Ocean. Northwest Atlantic Leatherback Working Group. (2018). Northwest Atlantic leatherback turtle (*Dermochelys coriacea*) status assessment (Bryan Wallace and Karen Eckert, Compilers and Editors). (WIDECAS Technical Report No. 16).
- Wallace, B. P., Tiwari, M., & Giron-dot, M. (2013). *Dermochelys coriacea* (The IUCN Red List of Threatened Species 2013: E.T6494A43526147).

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